AI Demand Forecasting Dashboard

Chart-by-chart explanation

Overview

This dashboard analyses historical product demand (Jan 2023 to Nov 2026) and uses a forecasting model to project demand for the next 12 months. It answers four questions: **How much have we sold? When does demand peak? Which products and categories drive it? What should we expect next year?**

Dashboard layout

Section	Visuals
Filters (top)	Year, Month, Category, Sub-Category, Product Name
KPI cards (row 1)	Total Historical Demand, Forecasted Demand (next 12 months), Average Monthly Demand, Number of Products, Forecasted Demand Growth %
Trend row (row 2)	Actual vs Forecasted Demand (line), Monthly Demand Trend (bar)
Breakdown row (row 3)	Top 10 Products, Bottom 10 Products, Category-wise Demand (donut), Product-wise Forecast (Top 10)

1. Filters (Slicers)

The five drop-downs at the top (**Year, Month, Category, Sub-Category, Product Name**) control every visual on the page. All are set to "All" by default, so the numbers shown are for the whole dataset.

- Select a Category (for example Technology) and every card and chart recalculates for that category only.
- Use Year and Month together to isolate a season, such as September 2025.
- Use Product Name to study one item, for example how a single stapler or chair is expected to perform.

2. KPI Cards

The cards give the headline numbers at a glance before the detailed charts.

2.1 Total Historical Demand: 35K

The total number of units demanded across all products from Jan 2023 to the latest month. It is the baseline size of the business. Because it responds to the filters, it also tells you how big any selected category, product or year is.

2.2 Forecasted Demand (Next 12 Months): 14,011 units

The total demand the model expects over the coming 12 months. The title is partly cut off in the screenshot (it shows only "Months)"), but the value matches the forecast line in the main chart and the +17.3% growth card, so it is the next-12-month forecast. Use it as the planning figure for purchasing, inventory and production.

2.3 Average Monthly Demand: 747

Total historical demand divided by the number of months of history, which is roughly 747 units a month. It is the "normal" month. Months well above it (September, November, December) are peak months; months below it (January, February) are slow months.

2.4 Number of Products: 1,843

The count of distinct products (SKUs) with demand. With 35K units spread over 1,843 products, the average product has only about 19 units of lifetime demand. That means most products sell in small quantities and a few products carry a large share, which the Top 10 / Bottom 10 charts confirm.

2.5 Forecasted Demand Growth %: +17.3%

The percentage change between the next 12 months of forecast demand and the last 12 months of actual demand. A positive value (green) means the model expects demand to grow. At +17.3%, demand is projected to be noticeably higher than the past year, so stock, staffing and supplier capacity should be planned for growth.

3. Actual vs Forecasted Demand (Line Chart)

This is the central chart of the dashboard. It compares what really happened with what the model predicts, month by month.

- **Blue solid line (Actual):** real monthly demand in units, Jan 2023 onward.

- **Orange dashed line (Forecast, fit on history):** the model's predictions for months that already happened. When it hugs the blue line, the model is accurate.
- **Yellow line in the shaded area (Forecast, next 12 months):** the future projection, roughly Dec 2026 to Nov 2027.
- **"Nov-26 partial month" marker:** November 2026 is not complete, so its actual value is understated and should not be read as a real drop.

Key insights

- **Strong seasonality:** every year demand is low in January and February (about 150 to 600 units) and rises sharply from September to December, with peaks around 1,000 to 1,700 units.
- **Upward trend:** each year's peak is higher than the previous one, which supports the +17.3% growth figure.
- **Model quality:** the orange fit follows the blue line's ups and downs closely, so the model has learned the seasonal pattern.
- **Forecast shape:** the next 12 months start high (about 1,600), dip to roughly 650 to 700 in the slow season, then climb to around 1,800 by the end of the horizon.

Business use

Build inventory ahead of the autumn peak and avoid over-stocking in the early-year trough.

4. Monthly Demand Trend (Bar Chart)

This bar chart adds up demand by calendar month (1 = January to 12 = December) across all years, so it shows the seasonal pattern clearly without the year-to-year noise.

Month	1	2	3	4	5	6	7	8	9	10	11	12
Units	1.6K	1.1K	2.6K	2.5K	2.9K	2.7K	2.7K	2.8K	5.1K	3.2K	4.1K	3.9K

- **September is the biggest month** at 5.1K, about 14% of all demand.
- **September to December together are about 16.3K units, close to half (roughly 46%)** of total demand. This is the peak season.
- **January and February are the weakest** (1.6K and 1.1K).
- March to August is steady at about 2.5K to 2.9K per month.
- Note: November includes the incomplete Nov-2026 month, so it is slightly understated.

Business use

Schedule promotions, hiring and stock replenishment around the March step-up and the September to December surge.

5. Top 10 Products by Demand (Green Bars)

A ranked list of the products with the highest total units demanded. The list is scrollable and long product names are truncated.

- Leaders: **Staples (218)**, **Staple envelope (166)** and **Easy-staple paper (139)**.
- Next come KI Adjustable-Height Table (76), Staples in misc. colors (76), Avery Non-Stick (71), Situations (62), Dana Halo (60) and iKross Bluetooth (60).
- **Insight:** demand is concentrated. The top three products alone account for 523 units, more than the next six combined, and they are all everyday consumables (staples and paper). These items need to stay in stock at all times.

6. Bottom 10 Products by Demand (Red Bars)

The products with the lowest demand, shown in red as a warning colour.

- Six products have only **1 unit** of demand (for example Binney & Smith, Boston 190, Global Enterprise, Motorola, Penpower, Xerox 20).
- Three more have **2 units** (3D Systems, Avery 5, Avery Hi-Lite).
- **Insight:** there is a very long tail of slow movers. These items tie up shelf space and capital for almost no sales and are candidates for delisting, bundling, discounting or make-to-order handling.
- Forecasts for products with so little history are unreliable, so treat them with caution.

7. Category-wise Demand (Donut Chart)

Shows how total demand splits across the product categories.

- **Office Supplies: 60.25%** (orange), the dominant category.
- **Technology: 18.15%** (light blue).
- **Furniture:** the remainder, about 21.6% (dark blue). Its label is cut off in the screenshot, so this figure is calculated as 100% minus the other two.
- **Insight:** about three of every five units come from Office Supplies, which fits the staples and paper leading the Top 10. Furniture and Technology are lower in volume, but each unit is likely higher in value, so check revenue as well as units before prioritising categories.

8. Product-wise Forecast, Top 10 (Clustered Bar Chart)

Compares, for the ten products with the highest forecast, **last 12 months actual** (blue) against **next 12 months forecast** (orange). The number at the end of each bar is the forecast in units.

Product	Next 12 mo forecast
Easy-staple paper	66
Staples	65
Staples in misc. colors	55
Staple envelope	50
Global Wood Trimmed Manager's Task Chair	38
KI Adjustable-Height Table	35
GBC Instant Report Kit	35
Lesro Sheffield Collection	34
Honeywell Enviracaire Purifier	34
Situations Contoured Folding Chair	33

- For most products the orange bar is longer than the blue bar, meaning growth is expected, consistent with the +17.3% overall figure. For Staples the two bars are nearly equal, so demand is expected to stay flat.
- This list differs from the historical Top 10 because it ranks by **expected** demand in the next year, not lifetime demand. Furniture and other items (task chairs, table, purifier) appear here.
- **Use:** this is the shortlist for next year's purchase planning and supplier negotiations.

9. Summary of Insights and Recommendations

- Demand is growing: the forecast is **+17.3%** (14,011 units over the next 12 months).
- Demand is highly seasonal: **September to December** is about 46% of annual volume, and January and February are slowest.
- Office Supplies, especially staples and paper, is the core of the business (about 60% of units).
- A long tail of products sells only 1 to 2 units; review them for rationalisation.
- Plan stock and promotions around the seasonal peak, and prioritise the top forecast products listed in section 8.

Notes and limitations

- November 2026 is a partial month, so recent actuals look lower than they really are.
- Forecasts are statistical estimates. Accuracy is best at the overall level and weaker for low-volume products.
- Two labels in the screenshot are truncated (the second KPI card title and the Furniture percentage); the interpretations above are inferred from the surrounding numbers.